

SALLA VARRIOTUNTURI, Finland

WMO: 028190

Lat: 67.749N

Lon: 29.611E

Elev: 360

StdP: 97.07

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.5	(c) -21.1	(d) -26.2	(e) 0.4	(f) -23.4	(g) -23.6	(h) 0.5	(i) -20.9	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.5	(c) 24.1	(d) 15.5	(e) 21.8	(f) 14.8	(g) 19.8	(h) 14.0	(i) 17.1	(j) 21.6	(k) 15.8	(l) 20.5	(m) 14.6	(n) 18.4	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.3	18.8	13.9	10.3	17.1	12.8	9.6	16.0	49.5	21.6	45.5	20.6	42.0	18.4	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -26.6	(f) 25.5	(g) 2.7	(h) 2.6	(i) -28.6	(j) 27.4	(k) -30.1	(l) 28.9	(m) -31.6	(n) 30.4	(o) -33.6	(p) 32.2		
(4) N/A	(5) N/A	(6) N/A	(e) -26.4	(f) 18.8	(g) 2.9	(h) 1.9	(i) -28.5	(j) 20.2	(k) -30.2	(l) 21.3	(m) -31.8	(n) 22.4	(o) -33.9	(p) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-11.8	-10.4	-6.7	-1.2	5.0	9.5	13.6	11.3	6.8	-0.4	-5.4	-7.9		
	DBStd	9.57	5.20	5.73	4.22	3.81	4.79	4.74	4.45	3.95	3.53	3.66	4.23	4.81		
	HDD10.0	3831	674	572	519	337	170	65	16	32	106	322	462	556		
	HDD18.3	6616	933	806	777	587	414	267	157	222	346	580	712	814		
	CDD10.0	273	0	0	0	0	14	51	127	71	10	0	0	0		
	CDD18.3	17	0	0	0	0	1	3	10	3	0	0	0	0		
	CDH23.3	95	0	0	0	0	2	19	58	16	0	0	0	0		
	CDH26.7	11	0	0	0	0	0	1	7	3	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	597	38	34	31	35	49	67	80	67	58	56	41	41		
	PrecMax	751	70	81	61	75	76	118	130	126	105	117	80	71		
	PrecMin	450	13	12	7	12	25	14	36	16	16	18	10	9		
	PrecStd	81	14	17	15	15	15	31	27	30	25	25	17	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	1.5	4.1	12.5	22.6	25.8	27.8	25.5	16.9	8.9	3.7	1.2		
		MCWB	-0.6	1.0	1.3	6.8	12.1	15.8	17.3	17.6	12.2	8.5	3.3	0.6		
	2%	DB	-2.0	-0.1	1.7	7.6	18.9	22.5	24.8	21.7	14.9	6.7	2.2	0.0		
		MCWB	-2.3	-0.6	0.0	3.5	11.1	15.0	16.1	15.6	11.3	5.7	1.8	-0.4		
	5%	DB	-3.6	-1.3	0.2	5.6	15.7	19.4	22.7	19.6	13.4	5.6	0.8	-0.7		
		MCWB	-4.0	-1.7	-1.0	2.3	9.3	12.7	15.0	14.5	10.6	4.9	0.6	-1.0		
	10%	DB	-4.9	-2.7	-0.8	4.0	12.6	17.0	20.7	17.4	11.9	4.4	0.1	-1.8		
		MCWB	-5.3	-3.1	-1.7	1.3	7.6	11.2	14.7	13.2	9.7	3.8	-0.1	-2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	1.1	1.6	6.3	13.0	17.2	19.2	18.7	13.6	8.6	3.6	0.8		
		MCDB	-0.4	1.4	3.3	11.9	19.6	23.5	23.1	24.7	15.4	8.7	3.7	1.0		
	2%	WB	-2.3	-0.4	0.3	4.0	11.5	15.7	17.6	16.4	12.0	6.0	1.7	-0.3		
		MCDB	-2.0	-0.1	1.4	6.8	18.2	21.2	21.9	20.4	13.9	6.4	2.0	0.0		
	5%	WB	-3.9	-1.6	-0.7	2.6	9.8	13.6	16.4	14.7	11.0	4.9	0.6	-1.0		
		MCDB	-3.6	-1.3	0.0	5.0	15.1	17.6	20.9	18.4	13.0	5.5	0.8	-0.7		
	10%	WB	-5.2	-3.0	-1.7	1.8	8.0	12.2	15.3	13.8	10.0	3.9	-0.1	-2.1		
		MCDB	-4.9	-2.7	-0.9	3.6	11.9	16.1	19.7	16.5	11.7	4.3	0.1	-1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	4.8	5.7	6.5	7.4	7.6	7.5	6.7	5.2	3.4	3.9	4.3		
		MCDBR	5.7	4.7	5.2	7.7	12.1	10.1	10.1	9.1	7.0	4.1	3.9	3.8		
		MCWBR	5.7	4.7	4.3	4.9	6.5	5.1	4.5	4.7	4.4	3.7	3.9	3.8		
		MCDBR	5.6	4.6	4.7	6.8	11.4	9.0	8.8	8.1	6.1	3.8	3.9	3.7		
	5% WB	MCWBR	5.6	4.6	4.1	4.4	6.5	5.3	4.6	4.7	4.4	3.6	3.9	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.190	0.248	0.288	0.317	0.337	0.338	0.352	0.333	0.304	0.268	0.354	N/A		
	taud		1.913	2.080	2.147	2.192	2.387	2.467	2.448	2.475	2.534	2.299	2.748	N/A		
	Ebn at Noon		121	554	744	828	857	867	839	817	758	544	166	0		
	Edh at Noon		10	49	82	105	98	93	92	80	59	39	13	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.51	1.95	3.64	4.68	4.95	4.51	3.23	1.72	0.55	0.10	0.00		
	RadStd		0.00	0.07	0.15	0.23	0.47	0.41	0.51	0.31	0.15	0.09	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page